



UNIVERSITY OF CAPE TOWN

Financial Analysis and Portfolio Management

Final Examination - 19 October 2019

Portfolio Optimization: ECO 4102Z

3 Hours

Maximum Marks: 100

INSTRUCTIONS TO CANDIDATES

1. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
2. Mark allocations are shown in brackets [].
3. Attempt all 7 questions, beginning your answer to each question on a new page.
4. Show all calculations where this is appropriate.
5. Write legibly using blue or black ink.

Additional Material: Formulae Sheet

1. Define the following terms:

- (a) State-by-state dominance [2 marks]
 (b) Second-order stochastic dominance [2 marks]

Consider the following investments whose payoffs, in millions of rand, have the following probability distribution(s):

Investment 1		Investment 2	
Payoff	Probability	Payoff	Probability
R1m	0.25	R3m	0.33
R7m	0.50	R5m	0.33
R12m	0.25	R8m	0.34

- (c) If the only available choice is 100% of your wealth in Investment 1 or 100% in Investment 2 and you choose on the basis of mean only, which investment is preferred. [3 marks]
 (d) Compare the two investments using the second-order stochastic dominance criterion. [5 marks]
 (e) Explain the practical limitation(s) of the concept of stochastic dominance in modelling investor choice. [2 marks]

[Total: 14 marks]

2. (a) In the context of mean-variance portfolio theory, define the term ‘efficient portfolio’. [2 marks]
 (b) Suppose that there are two efficient portfolios: A and B , with the following properties.

	Expected Return	Variance
Asset A	0.08	0.2
Asset B	0.10	0.4

The risk-free rate is 5 percent, and the correlation between the two portfolios is -0.2. Determine the expected return and standard deviation of the portfolio of A and B that maximizes the Sharpe ratio (i.e. the optimal portfolio). [10 marks]

- (c) Suppose your aunt, an engineer with very limited understanding of finance and investments, is considering investing in a well known index fund with a variance of 0.08 and an expected return of 0.09. She has approached you for investment advice. Based on your calculations in (b) above, advise her on the desirability of this fund [3 marks]
 [Total: 15 marks]

3. Consider a market in which the Capital Asset Pricing Model (CAPM) holds. There are two assets with the following attributes.

Rate of return (per annum)				Variance/Covariance Matrix		
State	Probability	Asset 1	Asset 2		Asset 1	Asset 2
1	0.2	5.00%	11.00%	Asset 1	0.00068	0.00102
2	0.3	10.00%	15.00%	Asset 2	0.00102	0.00181
3	0.1	8.00%	12.00%			
4	0.4	4.00%	5.00%			
Market capitalisation		40,000	60,000			

(a) Calculate the beta of each security. [8 marks]

(b) Determine the value of the risk-free rate of interest which is consistent with the results obtained in part (a), under the assumption that the CAPM holds. [5 marks]

[Total: 13 marks]

4. (a) Discuss the advantages of arbitrage pricing theory (APT) over the capital asset pricing model (CAPM) relative to diversified portfolios. [5 marks]

Consider the following information about two stocks (X and Y) and two common risk factors (1 and 2):

Stock	b_{i1}	b_{i2}	$E(R_i)$
X	1.25	3.2	14.25%
Y	2.4	2.5	16.6%

(b) Assuming that the risk-free rate is 6.5%, calculate the levels of the factor risk premia that are consistent with the reported values for the factor betas and the expected returns for the two stocks. [6 marks]

(c) You expect that in one year the stock prices for X and Y will be R600 and R450, respectively, none of the companies is expected to pay a dividend over the next year. Determine the expected current price of each company. [4 marks]

[Total 15 marks]

5. (a) Describe how domestic bank shares are expected to perform relative to consumer non-durable goods shares during:

- low economic growth
- high economic growth

[5 marks]

A retired professional is deciding whether to invest part of his portfolio in a number of domestic bank shares. His requirement is for capital growth over a short horizon of one to two years.

Over the past year bank shares have fallen in value by an average of 60% due to concerns about global economic growth. The current forecast is for an economic recovery in three to four years' time.

(b) Explain why the proposed investment might not be suitable for the individual. [6 marks]

[Total 11 marks]

6. (a) Suppose due to recent economic developments in the country, you expect technology stocks, such as software development companies and hardware equipment firms, to perform well over the next three to six months. As an active portfolio manager, describe how would you take advantage of this forecast. [2 marks]
- (b) Discuss why it is important for investors to perform style analysis on managers. [5 marks]
- (c) Describe conditions under which Value stocks might be expected to outperform Growth stocks. [4 marks]
- [Total: 11 marks]
7. Suppose a risk averse charity based in Gugulethu, Capetown, has a small equity portfolio. The trustees are considering the award of a new mandate to an active equity manager to generate higher returns. The charity has short listed four managers as part of their selection process. They have been provided with the following information on performance over the last four quarters:

Manager	Fund return				Covariance with index
	Q1	Q2	Q3	Q4	
A	2.5%	1.5%	4.5%	6.0%	0.5
B	1.5%	-2.5%	5.0%	4.5%	0.3
C	6.0%	3.5%	2.5%	10.5%	0.74
D	3.5%	4.0%	4.0%	6.0%	0.35
Index benchmark return	2.0%	2.0%	3.0%	3.0%	
Standard deviation of index	0.7				
Risk-free rate	4% p.a.				

- (a) Assess the outperformance over the last year of each investment manager relative to the benchmark, and rank the performance of the managers. [6 marks]
- (b) Calculate the risk adjusted performance of the four investment managers using the Jensen measure (or Jensen's alpha). Based on this measure, which manager performed well? [9 marks]
- (c) Assess which manager is likely to be the most suitable for the investor. [3 marks]
- (d) Explain the circumstances in which standard deviation is a more appropriate measure of risk for an investor than beta. [3 marks]

[Total: 21 marks]